

# CMS vs. Segmentation & Sampling Service

## Feature Delta Analysis

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**Purpose:** Identify capabilities proposed in the Panacea Cohort Management System (CMS) PRFAQ that are not covered by the Minerva Population Sampling and Aggregation functional design — to inform what must be accounted for if the two systems converge into a single service.

### Context

Two documents propose new infrastructure for managing device populations:

**1. Panacea CMS PRFAQ** — A purpose-built service for tracking groups of devices affected by detected problems, enabling troubleshooting workflows across Anomaly Detection, Issue Detection, Case Management, and Customer Support.

**2. Minerva Sampling & Aggregation** — A system for reducing telemetry footprint by 90% through confidence-based population stratification sampling, enforced pre-storage on devices.

Both systems share a common foundation: grouping devices by shared characteristics (device type, software version, region) and managing membership in those groups. The question is whether they should be one service or two.

### Delta: CMS Capabilities Not in Segmentation Doc

#### 1. Problem-Driven Cohorts (Affected/Control Paradigm)

**CMS:** Groups devices for troubleshooting — tracking which devices exhibit a detected problem. Cohorts are typed as affected, control, composite, or investigation-sample.

**Segmentation:** Groups devices for telemetry optimization only. No concept of affected vs. control or problem-linked grouping.

**Action:** Add a cohort type taxonomy (affected, control, composite, investigation-sample) alongside sampling strata.

#### 2. Temporal Membership Tracking (Observation Windows)

**CMS:** Each membership record contains DSN, immutable first\_seen timestamp, occurrence\_count, and a bounded set of recent non-overlapping observation time ranges. Teams can distinguish a device affected once from one affected repeatedly.

**Segmentation:** Membership is binary — in sample or out with a rotation TTL (3-7 days). No temporal history.

**Action:** Extend membership record schema to support both models — simple TTL for sampling, temporal observation windows for troubleshooting.

### 3. DSN-to-Cohort Reverse Lookup (CS Path)

**CMS:** A single API call resolves a DSN to every cohort it belongs to, along with linked Issues, in under 100ms at p99. Powers CS associate workflows and AI self-service.

**Segmentation:** Has 100ms forward lookup (device -> sampling config). No reverse lookup (DSN -> all problems/cohorts).

**Action: Add a materialized reverse index (DSN -> all cohort IDs + linked Issues) with O(1) lookup and new LookupDSN API.**

### 4. Cross-Domain Cohort Correlation Keys

**CMS:** Every cohort has a deterministic correlation key — a hash of device family, firmware version, and region. Two independent systems resolve to the same key without coordination.

**Segmentation:** Strata defined by system owners for sampling. No deterministic cross-system correlation mechanism.

**Action: Add deterministic correlation key generation so independent systems can recognize overlapping populations.**

### 5. Composite Cohorts (Hierarchical Composition)

**CMS:** Composite cohorts are dynamic unions of child cohorts, evaluated at query time. Supports 3 levels: Anomaly -> Issue -> Case.

**Segmentation:** Strata are flat groupings. No hierarchy or composition.

**Action: Add dynamic composition engine supporting union-based composite groups up to 3 levels deep.**

### 6. Cohort Lifecycle Management

**CMS:** Full lifecycle: active -> resolved -> archived. Staleness threshold (60 days) triggers archival. Archived cohorts can be reactivated. DSN records purged after 12 months per Policy 97.

**Segmentation:** No lifecycle states. Strata exist as long as population definition exists.

**Action: Add lifecycle state machine with configurable staleness thresholds, archival policies, reactivation, and data retention rules.**

### 7. Control Group Management with Temporal Exclusion

**CMS:** APIs for control group creation with matching criteria. Enforces temporal exclusion — device cannot appear in both affected and control cohorts for overlapping time windows. Automatic replacement if control device becomes affected.

**Segmentation:** No control group concept. Sampling is about statistical representation, not differential analysis.

**Action: Add control group creation APIs, temporal exclusion invariant enforcement, and automatic replacement logic.**

## 8. Cross-Cohort Set Operations

**CMS:** Query API supports intersection, union, and difference. Cross-cohort queries return within 2 seconds at p95.

**Segmentation:** No set operations across strata or samples.

**Action:** Add cross-group query APIs supporting intersection, union, difference with <2s p95 latency.

## 9. Linked Entity References

**CMS:** Cohort metadata includes linked entity references — Issue IDs, Case IDs, Anomaly IDs. Enables navigation from cohort to investigation context.

**Segmentation:** No concept of linking to external entities.

**Action:** Extend metadata model to support optional linked entity references (Issue ID, Case ID, Anomaly ID).

## 10. Dual Ingestion Modes (Real-time + Batch)

**CMS:** Real-time event bus ingestion (processed within seconds) + Batch FULL\_SYNC reconciliation. Both feed the same membership index and are not mutually exclusive.

**Segmentation:** Periodic batch-driven sample generation only.

**Action:** Add real-time event ingestion path alongside existing batch recalculation, both feeding same index.

## 11. Explicit Device Removal with Audit Trail

**CMS:** Three removal mechanisms — explicit API removal, lifecycle-based removal, batch reconciliation. All removals recorded with timestamp and reason.

**Segmentation:** Devices leave samples only via TTL expiry or becoming non-sampleable. No explicit removal API.

**Action:** Add RemoveMember API with required reason field and audit logging.

## 12. Privacy Model (DSN Classification + pDSN)

**CMS:** DSNs classified as Highly Confidential / PII per Policy 97. Encrypted at rest and in transit. Privacy review planned for Q2 2026 to evaluate pDSN pseudonymization.

**Segmentation:** Does not explicitly address DSN data classification or Policy 97 compliance.

**Action:** Adopt CMS privacy model. Coordinate single privacy review covering both use cases.

## 13. Access Control Model (Write-Restricted, Read-Open)

**CMS:** Write access restricted to authorized Panacea subsystems via CloudAuth. Read access open to any valid credential. Human DSN access requires L7+ approval + Device Privacy sign-off.

**Segmentation:** Simple system-owner model. No tiered access control.

**Action:** Implement CloudAuth-based write restrictions, tiered read access, and human-access controls for DSN data.

## Summary Table

| #  | CMS Capability                      | Segmentation Equivalent     | Action for Unified Service                   |
|----|-------------------------------------|-----------------------------|----------------------------------------------|
| 1  | Affected/Control cohort types       | None (only sampling strata) | Add cohort type taxonomy                     |
| 2  | Temporal observation windows        | Binary in/out + TTL         | Extend membership record schema              |
| 3  | DSN -> all-cohorts reverse lookup   | Forward lookup only         | Add materialized reverse index + API         |
| 4  | Deterministic correlation keys      | None                        | Add correlation key generation               |
| 5  | Composite cohorts (3-level)         | Flat strata                 | Add dynamic composition engine               |
| 6  | Lifecycle states                    | None                        | Add lifecycle state machine                  |
| 7  | Control groups + temporal exclusion | None                        | Add control group APIs + exclusion invariant |
| 8  | Set operations                      | None                        | Add cross-group query APIs                   |
| 9  | Linked entity references            | None                        | Extend metadata model                        |
| 10 | Real-time event ingestion           | Batch-only                  | Add event bus ingestion path                 |
| 11 | Explicit removal + audit            | TTL/offline only            | Add RemoveMember API                         |
| 12 | DSN privacy + pDSN                  | Not addressed               | Add privacy controls + review                |
| 13 | Tiered access control               | Simple model                | Add write/read access tiers                  |

## Architectural Recommendation

The Segmentation doc's Population Strata Grouping Service is the natural home for the shared dimension model and stratification logic. The CMS capabilities would layer on top as a 'Cohort Management' domain within the unified service, sharing the population/strata infrastructure but adding troubleshooting-specific semantics.

### Suggested Unified Service Layers:

#### Layer 3: Domain-Specific APIs

- +- Sampling Domain: Sample configs, confidence eval, rotation, enforcement
- +- Troubleshooting Domain: Affected/Control mgmt, composite cohorts, DSN reverse lookup, lifecycle, set operations, linked entities

#### Layer 2: Shared Membership & Query Engine

- Membership index (DSN -> groups), dimension filtering, correlation keys, audit logging, real-time + batch ingestion, access control (CloudAuth)

#### Layer 1: Population Stratification Foundation

- Population definitions, strata computation (device type, FW, region), external source integration (DMS, Panorama), sampleable entity determination

## Next Steps

1. Align with Minerva team (Nathan/Adam/Nick) on whether the Population Strata Grouping Service design can absorb CMS membership and query requirements without compromising sampling latency targets.
2. Privacy review — Coordinate a single privacy review covering both use cases (sampling + troubleshooting) since both store DSN-level data.

3. API contract design — Draft a unified Smithy model supporting both sampling strata and troubleshooting cohorts as first-class group types with shared query infrastructure.
4. Phase alignment — CMS Phase 1 (Q2 2026) and Segmentation POC are on similar timelines. Determine if a shared Layer 1 can be delivered first, with domain-specific layers following.